

**AI Programming Foundations Capstone Project: Analyzing Airbnb Listings in New York
City**

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Abstract

Replace this text with your abstract. Summarize the research problem, method, main findings, and conclusion. Add keywords below when required by your institution.

Keywords: keyword one, keyword two, keyword three

Overview

This project implements a reproducible Python workflow for cleaning, exploring, and visualizing the New York City Airbnb Open Data dataset (Gomonov, 2019). The dataset is available from Kaggle at <https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data>. The analysis examines how listing prices and annual availability vary across New York City neighbourhood groups and room types. The complete workflow is contained in the accompanying `data_workflow.ipynb` notebook.

Dataset Description

The New York City Airbnb Open Data dataset contains listing-level information for Airbnb properties in New York City during 2019. The original CSV file contains 48,895 rows and 16 columns, with each row representing one listing.

The analysis focused primarily on `price`, `neighbourhood_group`, `room_type`, `availability_365`, `minimum_nights`, `number_of_reviews`, `reviews_per_month`, and `calculated_host_listings_count`. Latitude, longitude, listing identifiers, listing names, and host information were retained but were not central to the main comparisons.

The dataset includes the five neighbourhood groups Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. Its room-type categories are entire home or apartment, private room, and shared room.

References

Gomonov, D. (2019). *New york city airbnb open data* [Accessed 2026-08-04]. Kaggle. <https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data>

Supplementary Material

Place questionnaires, extended tables, source code excerpts, or other supporting material here when appropriate.